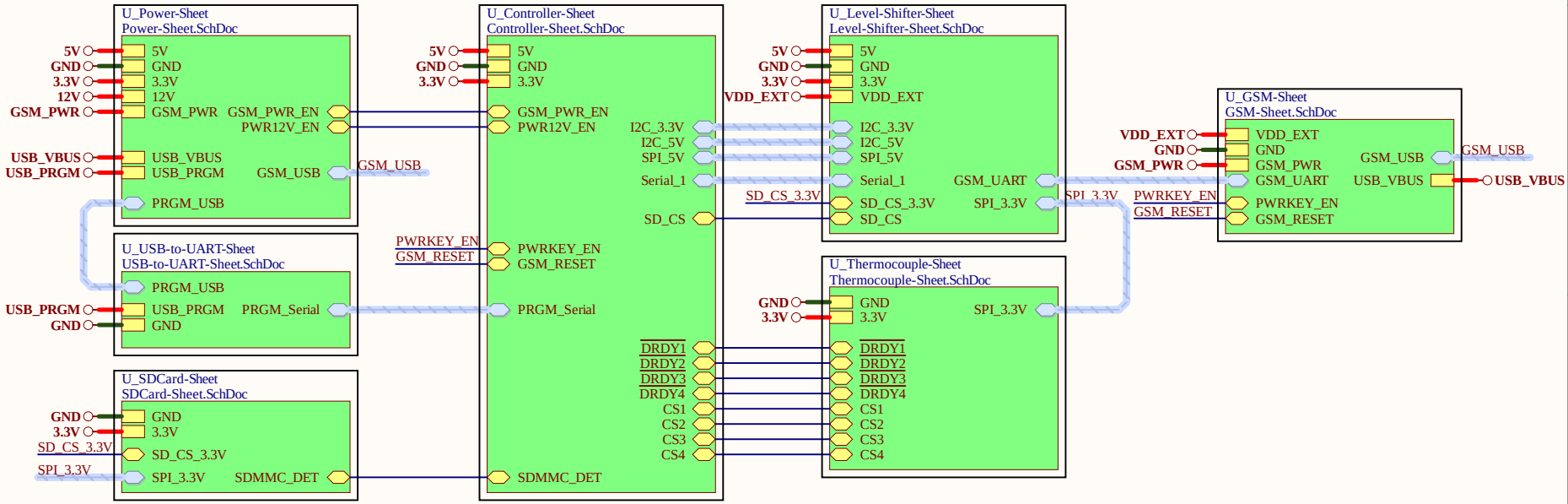
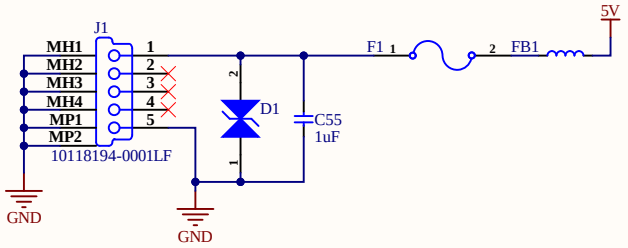


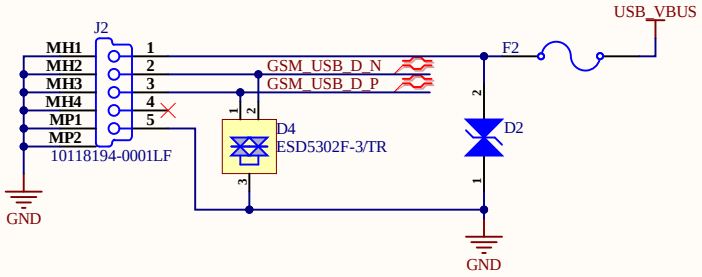
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File: Mega2560_Custom_v2.0.PrjPcb	
Date: 5/12/2023	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 8

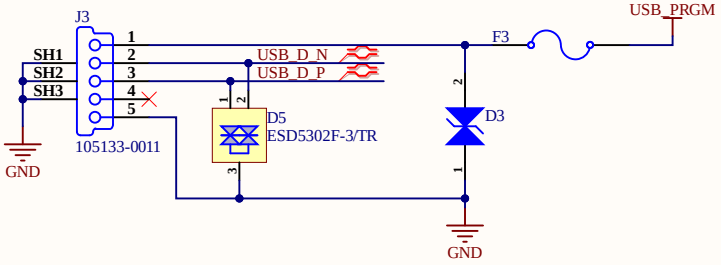
USB FOR BOARD POWER



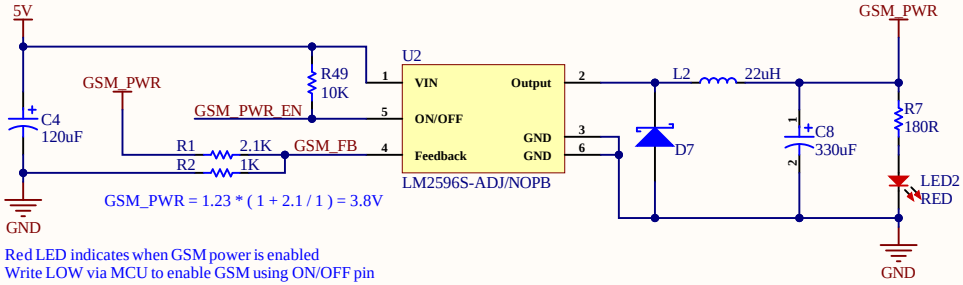
USB FOR GSM UPDATES



USB PROGRAMMING MEGA2560

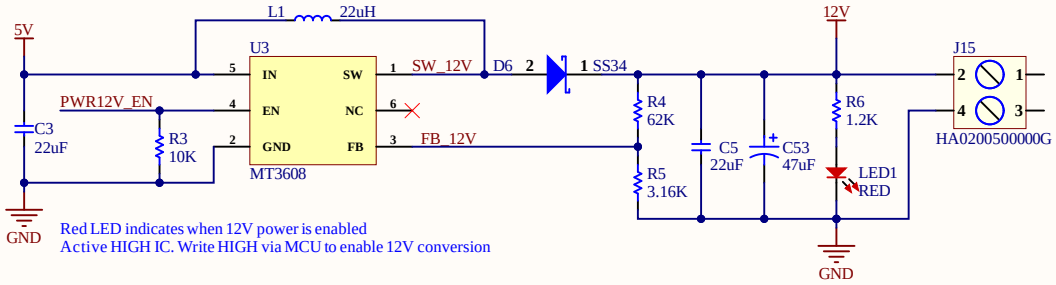


BUCK FOR GSM POWER 5V TO 3.8V



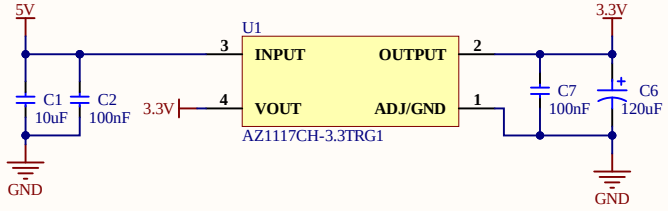
Red LED indicates when GSM power is enabled
Write LOW via MCU to enable GSM using ON/OFF pin

BOOST FOR WIKA SENSOR 5V TO 12V

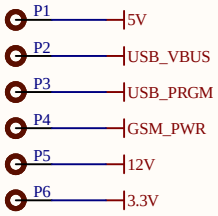


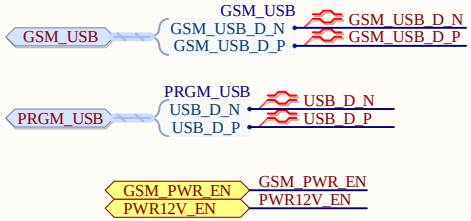
Red LED indicates when 12V power is enabled
Active HIGH IC. Write HIGH via MCU to enable 12V conversion

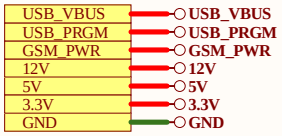
LDO 5V TO 3.3V



Test Points



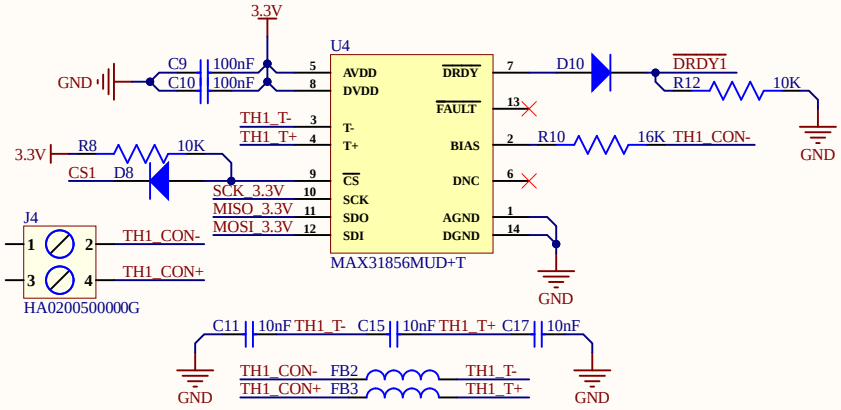




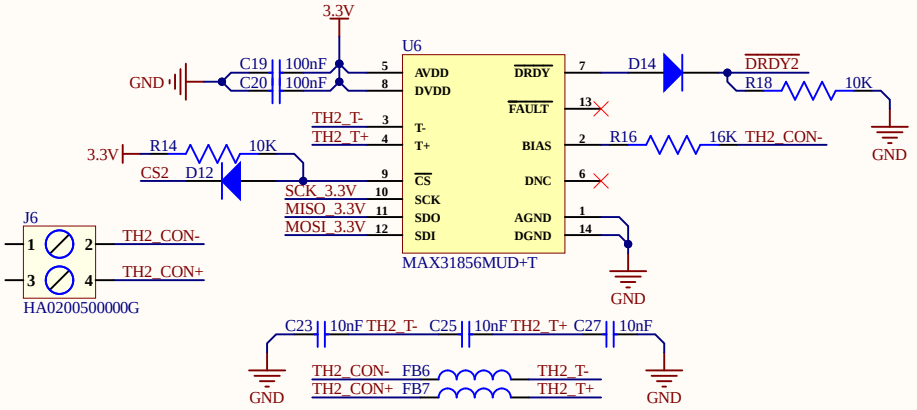
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D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg.
File does not exist.

Title: Power-Sheet.SchDoc	
File: Mega2560_Custom_v2.0.PrjPcb	
Date: 5/12/2023	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 8

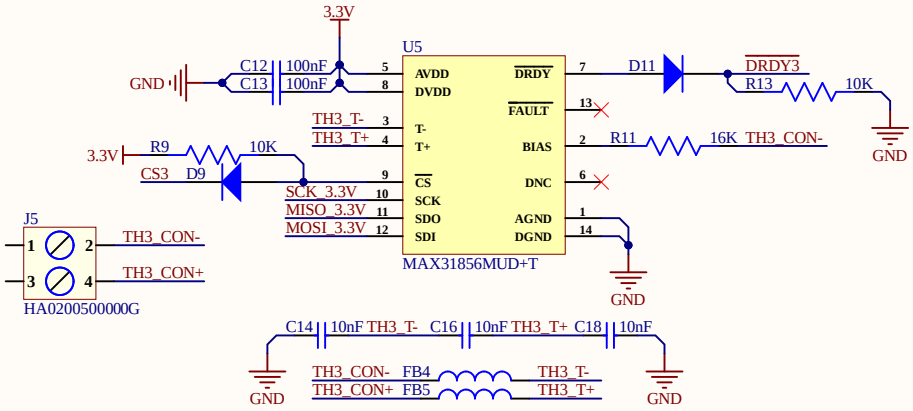
Thermocouple Input - 1



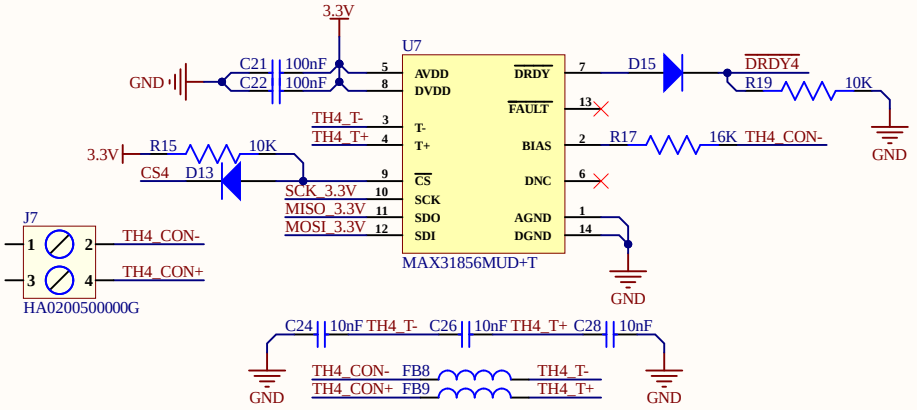
Thermocouple Input - 2



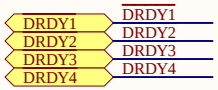
Thermocouple Input - 3

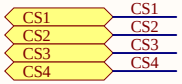


Thermocouple Input - 4







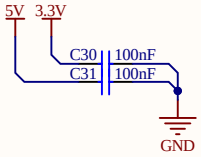


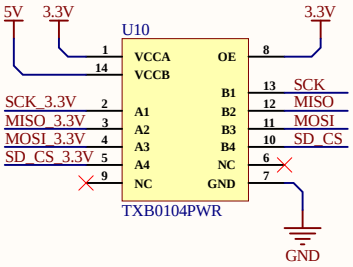


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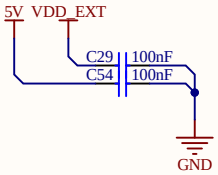
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Date: 5/12/2023	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 8

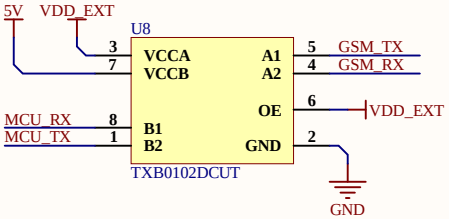
Level Shifter for SPI



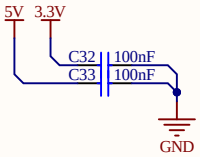


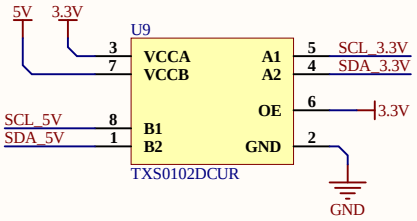
Level Shifter for Serial

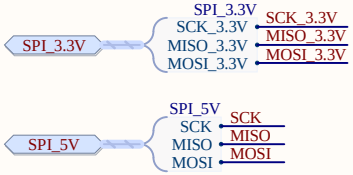


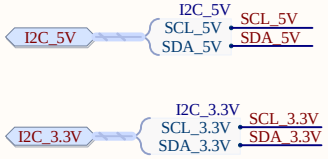


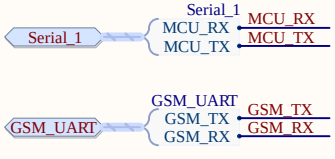
Level Shifter for I2C

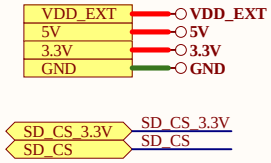








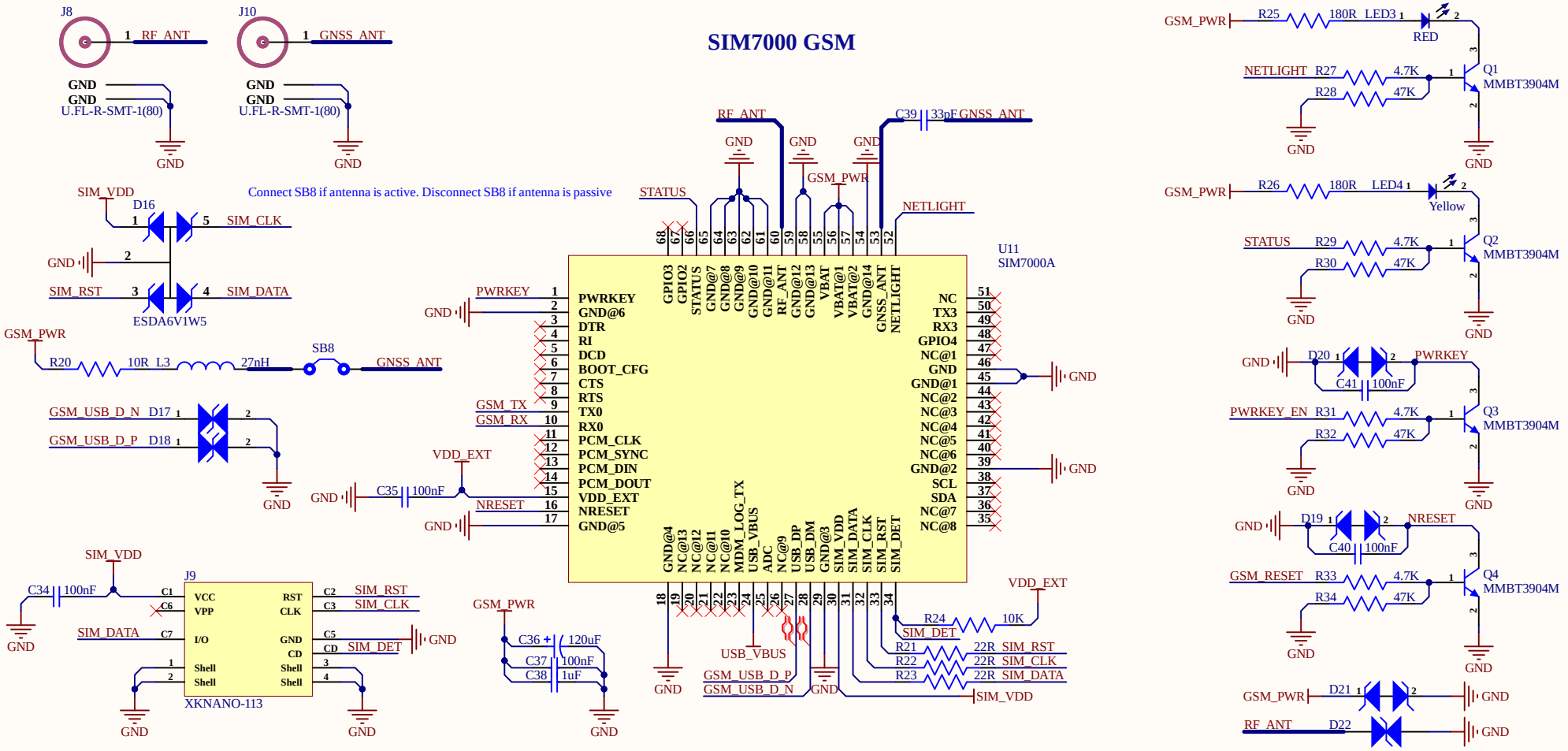




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Title: Level-Shifter-Sheet.SchDoc	
File: Mega2560_Custom_v2.0.PrjPcb	
Date: 5/12/2023	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 8

SIM7000 GSM



GSM_UART
GSM_TX
GSM_RX

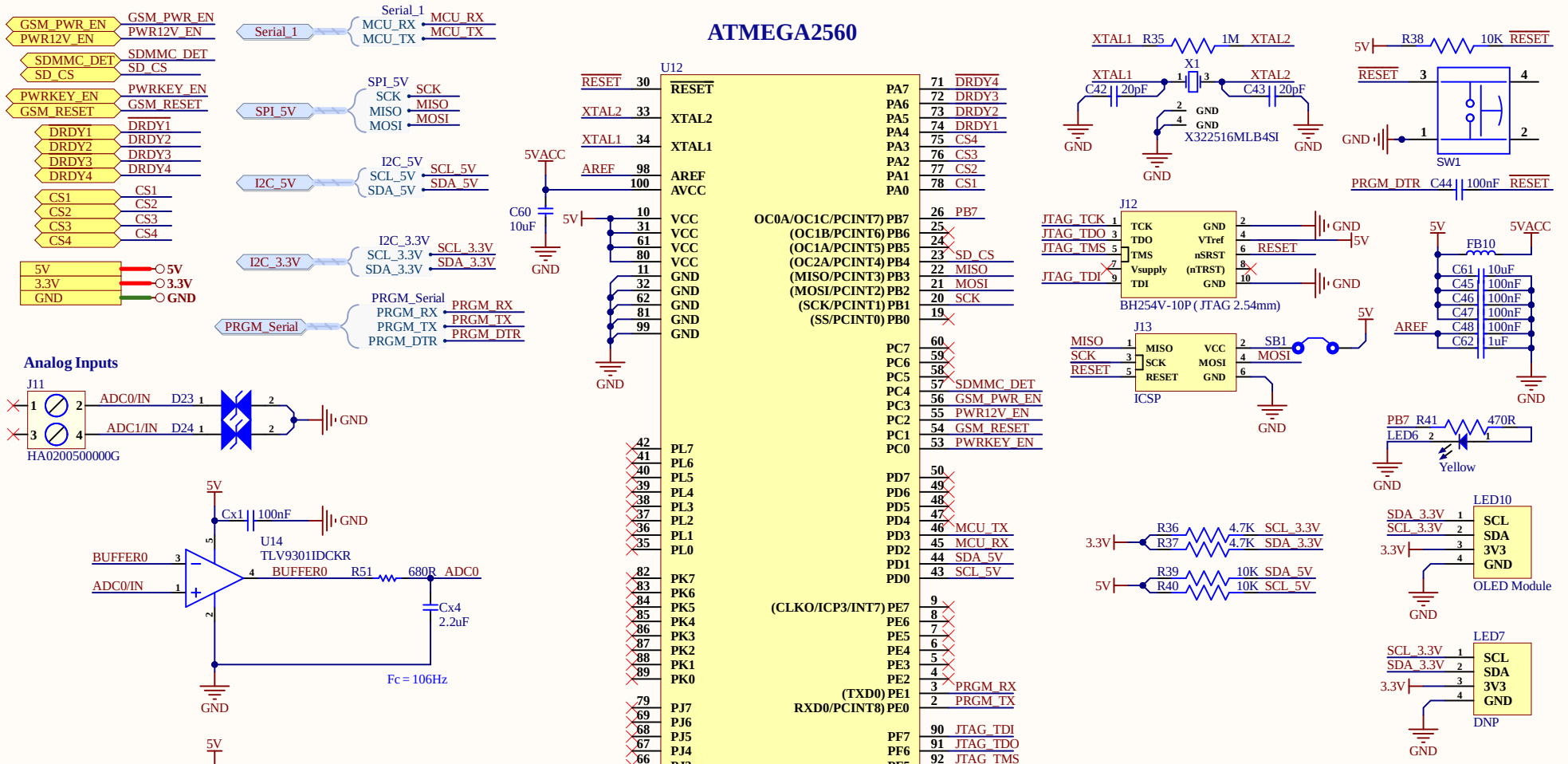
GSM_USB
GSM_USB_D_N
GSM_USB_D_P

PWRKEY_EN
GSM_RESET

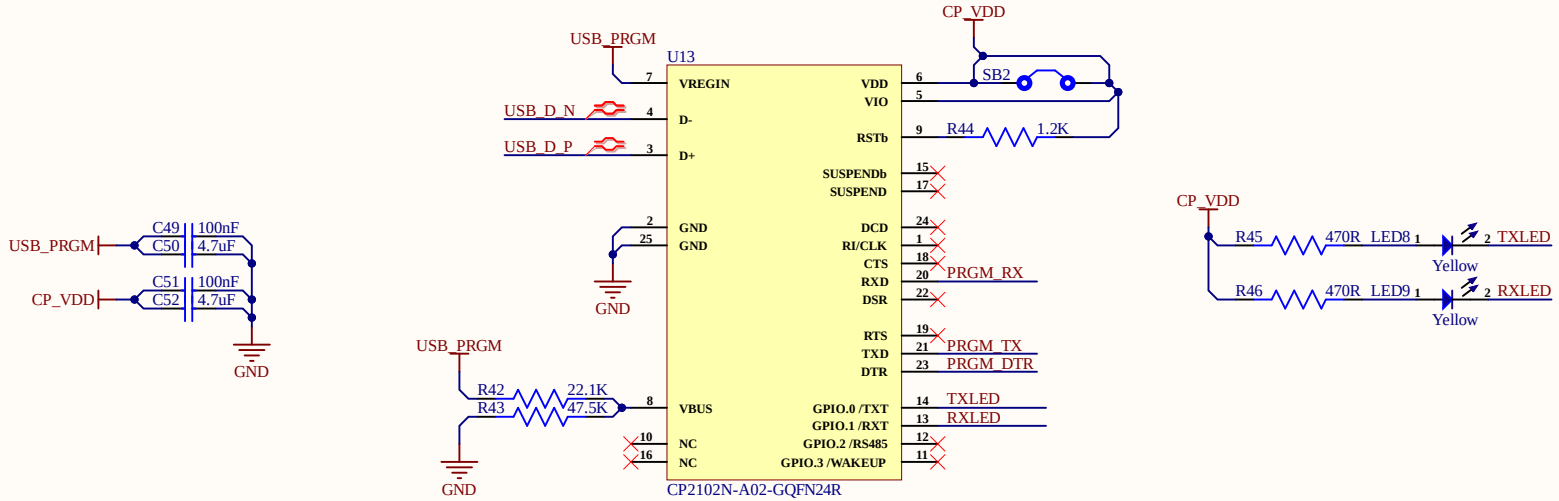
USB_VBUS
GSM_PWR
VDD_EXT
GND

Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: GSM-Sheet.SchDoc	
File: Mega2560_Custom_v2.0.PrjPcb	
Date: 5/12/2023	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 8


ATMEGA2560
Analog Inputs
J11
HA0200500000G
U12
ATMEGA2560-16AU
U14
TLV9301IDCKR
U15
TLV9301IDCKR
Fc = 106Hz
Fc = 106Hz
BH254V-10P (JTAG 2.54mm)
ICSP
SW1
X322516MLB4SI
OLED Module
DNP
Cannot open file
D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg.
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Title: Controller-Sheet.SchDoc
File: Mega2560_Custom_v2.0.PrjPcb
Date: 5/12/2023
Page Num: 6
Drawn By: Huzaifa Ali
Page Total: 8

USB-to-UART for Programming





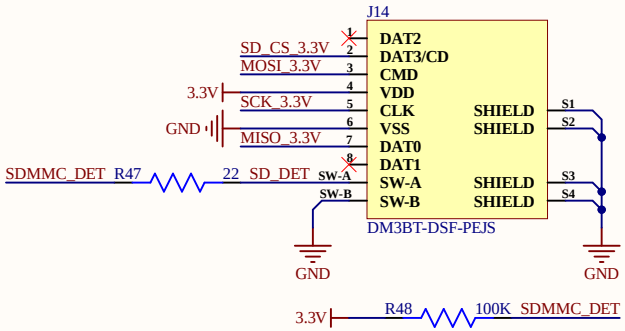


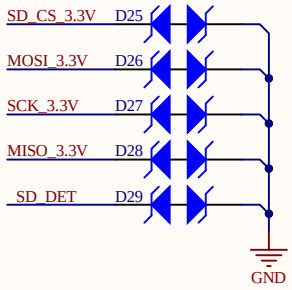


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Title: USB-to-UART-Sheet.SchDoc	
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Date: 5/12/2023	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 8

SD Card Interface





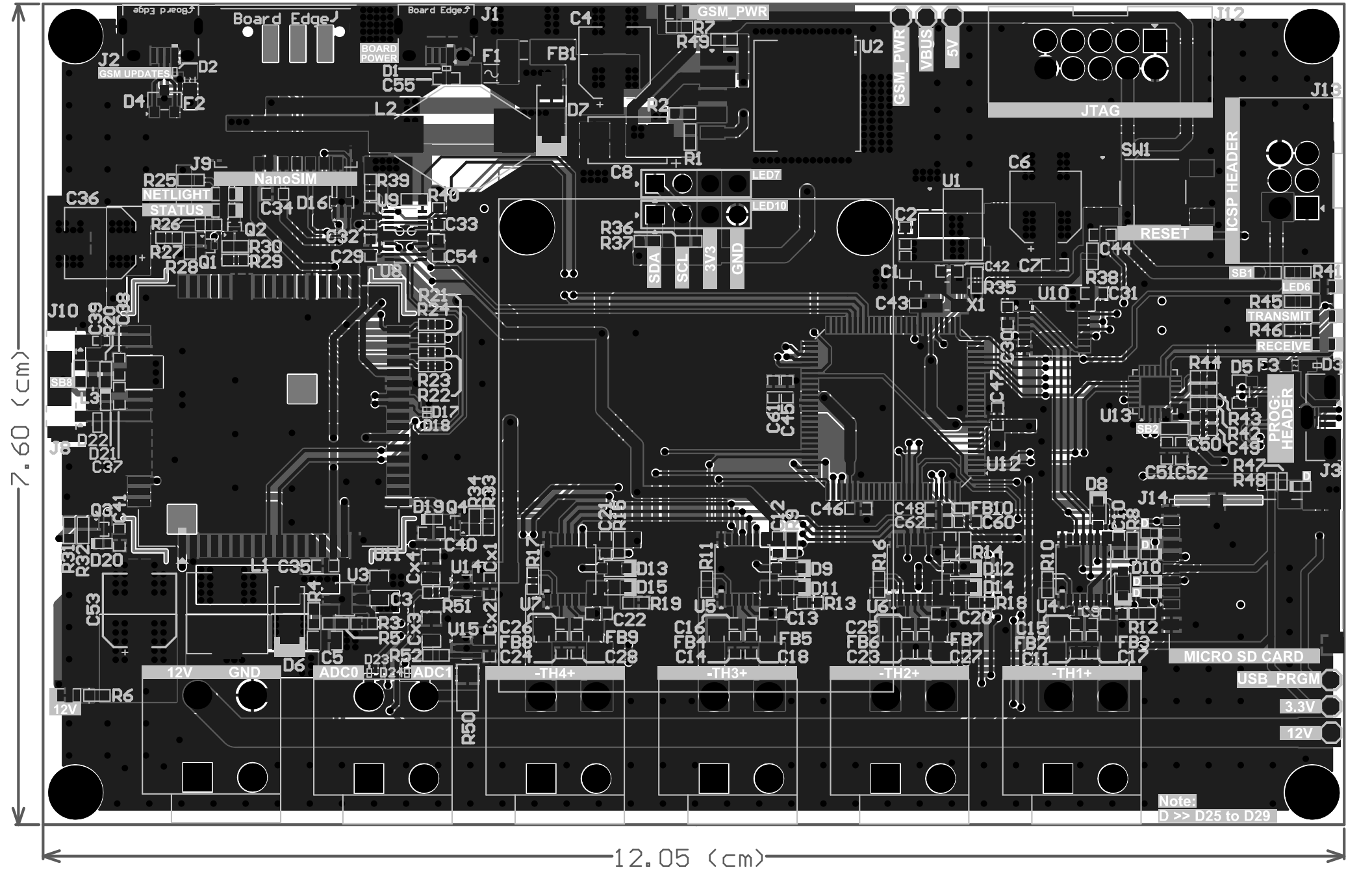






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Title: SDCard-Sheet.SchDoc	
File: Mega2560_Custom_v2.0.PrjPcb	
Date: 5/12/2023	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 8


7.60 (cm)
12.05 (cm)
Board Edge
J1
J2
GSM UPDATES
BOARD POWER
GSM_PWR
VBUS
5V
J12
JTAG
J13
ICSP HEADER
SW1
RESET
NanoSIM
J9
NETLIGHT
STATUS
LED7
LED10
SDA
SCL
3V3
GND
LED6
TRANSMIT
RECEIVE
PROG. HEADER
J3
J14
MICRO SD CARD
USB_PRGM
3.3V
12V
GND
ADC0
ADC1
-TH4+
-TH3+
-TH2+
-TH1+
Note:
D >> D25 to D29